Zero Trust Network Access (ZTNA) with Zscaler: Securing User-to-Application Connectivity

IMPLEMENT AND MANAGE ZERO TRUST NETWORK ACCESS (ZTNA) USING ZSCALER BY ENABLING SECURE, IDENTITY-BASED USER-TO-APPLICATION CONNECTIVITY WITHOUT EXPOSING INTERNAL NETWORKS.

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Project title: Implement and manage Zero Trust Network Access (ZTNA) using Zscaler by enabling secure, identity-based user-to-application connectivity without exposing internal networks.

AI-Enhanced User Protection (AIUP)

AI-Enhanced User Protection (AIUP) represents a paradigm shift in digital identity verification, employing artificial intelligence and machine learning to safeguard online platforms. It transcends conventional security models that depend on static elements like passwords or two-factor authentication (2FA). AIUP dynamically builds an evolving "trust score" for each user by continuously analyzing their behavioral patterns.

This framework is vital for contemporary online security as it overcomes the inherent weaknesses of traditional perimeter-based defenses. In an era where cybercriminals leverage sophisticated techniques such as credential theft, phishing, and social engineering, depending solely on "what you know" (passwords) or "what you have" (tokens) is inadequate. AIUP introduces a persistent, third layer of verification: "who you are," as defined by your unique interaction patterns. This approach empowers organizations to effectively counter account takeovers, minimize fraudulent activities, and guarantee that the individual interacting with a platform is authentic throughout their entire session.

Abstract

The inherent flaws of conventional VPN frameworks, especially their susceptibility to lateral movement and the provision of excessive network privileges, demand a fundamental move towards Zero Trust principles. This document outlines the holistic design, implementation, and administration of a Zero Trust Network Access (ZTNA) solution, utilizing Zscaler Private Access (ZPA) within a complex institutional environment. The core objective is to transition from broad network-level access to a model of application-specific, identity-focused connectivity.

Through the strategic placement of Zscaler App Connectors within the data center and the integration of the platform with established Identity Providers (like Azure AD) and endpoint protection mechanisms, the infrastructure guarantees that every user-to-application connection is facilitated by a cloud-native broker. This methodology effectively renders internal applications "invisible" from the internet, thereby shrinking the attack surface.

Additionally, the project delves into the synergy between Zscaler and the broader security ecosystem, incorporating SIEM platforms (such as QRadar) for consolidated logging and MISP for ingesting threat intelligence to facilitate adaptive policy decisions. The outcome is a resilient and scalable security posture that enforces least-privilege access, delivers exhaustive audit capabilities, and markedly improves the institution's defensive strength.

Use Case Scenarios

Scenario 1: Secure Remote Access for Faculty to ERP System

A faculty member working remotely requires access to the university's ERP system for grade submission. Through Zscaler Private Access (ZPA), a secure connection is established without placing the device on the internal network. The process verifies their identity via Azure AD MFA and evaluates their device's security posture before access is granted.

Output:

- User identity validated (Faculty_Group)
- Device posture confirmed (AV active, OS up-to-date)
- Encrypted micro-tunnel established exclusively to the ERP Application
- Underlying network remains completely concealed from the user

Scenario 2: External Researcher Accessing a Specific Database

An external research partner from another institution needs temporary, targeted access to a single research database hosted on VIT's servers. The administrator creates a policy in Zscaler that is both time-bound and application-specific. The researcher can then access only the authorized database, with no visibility into other areas of the VIT network.

Output:

- External user identity verified using temporary credentials
- Access policy enforced (Time-limited: 7 days, Application: Research-DB only)
- Database server remains hidden from internet-wide scans
- Session automatically ends upon policy expiration

Scenario 3: Student Access to Online Examination Portal

During the final exam period, a large number of students concurrently access the internal online examination portal from diverse locations. Zscaler ZPA ensures that only authenticated students with compliant devices can reach the exam server. The system mitigates potential DDoS attacks on the login page and prevents any efforts to access other internal systems.

Output:

- High-volume user authentication managed through cloud scalability
- Each student is connected only to the specific exam application
- Lateral movement attempts to the grade database or other servers are blocked
- Exam server IP address remains hidden from the public internet

Scenario 4: IT Administrator Performing Server Maintenance

An IT administrator needs to execute essential updates on an internal application server. Rather than using a VPN for full network access, they utilize ZPA to establish a direct connection to the server's dedicated management interface. The session undergoes real-time monitoring, and any unusual activity, such as an attempt to connect to unauthorized servers, is automatically terminated.

Output:

- Admin identity verified with MFA
- Direct micro-tunnel established solely to Server-Management-Interface
- All admin actions and commands are logged for auditing
- Any attempt to access another server is automatically blocked

Scenario 5: Mergers and Acquisitions - Temporary Contractor Access

A contractor brought in for a university system upgrade requires temporary access to a specific file server for data migration. Through Zscaler ZPA, the contractor receives application-specific access without being integrated into the university's Active Directory or receiving VPN credentials. This access is automatically revoked upon the project's completion.

Output:

- Contractor identity verified through a separate IdP integration
- Access granted exclusively to the File-Server-Migration share
- All file transfer activities are logged in Zscaler and the SIEM
- Access is automatically revoked after the 30-day project concludes

Stage 1: Web Application Security Assessment

Project Overview

This project focuses on creating an AI-enhanced identity verification system to bolster online security by analyzing and authenticating user identities through their behavioral patterns. It utilizes sophisticated machine learning methods to construct robust user profiles, effectively differentiating between authentic users and potentially harmful activities. This directly confronts the critical challenges of unauthorized access, transactional fraud, and cyber threats across diverse online platforms and services.

The primary goal is to substantially elevate online security by meticulously examining and validating user identities via their digital behavior. The system leverages cutting-edge machine learning approaches, including deep learning, neural networks, and natural language processing, to develop detailed behavioral profiles for each user. These profiles are dynamic, continuously adapting to the user's evolving behavior over time. A central feature is real-time

anomaly detection, which identifies deviations from established behavioral baselines and enables swift responses to mitigate potential security threats.

To further strengthen security, multi-factor verification is integrated, triggering supplementary verification steps when anomalies are identified. Transparent alerts for both users and administrators ensure clear communication and direction for resolving potential security incidents. Secure data storage and management are fundamental to safeguarding sensitive user profile information and anomaly data.

The architecture includes an integration layer designed for smooth interoperability with various online platforms and services. Comprehensive monitoring and reporting functionalities provide transparency and accountability for both performance and security. The design prioritizes scalability and load balancing to accommodate an expanding user base. Security and compliance are paramount, with measures such as data encryption in transit and at rest, deployment of firewalls and intrusion detection systems, and strict adherence to data protection and privacy regulations.

Leveraging cloud infrastructure is an optional but recommended path for enhanced flexibility and scalability. A commitment to the continuous training and refinement of machine learning models ensures the system remains effective against new and evolving threats and changing user behaviors. Geographic distribution of components can be employed to improve system redundancy and performance.

Vulnerability Assessment Tables

Practice Website: http://www.itsecgames.com/

S.No	Vulnerability Name	CWE Number
1	Improper Sanitization of File Paths	CWE-284: Improper Access Control
2	Invalid Certificate	CWE-395: Improper Certificate Validation
3	Missing X-Frame-Options Header	CWE-1021: Improper Restriction of Rendered UI Layers or Frames
4	X-Content-Type Options Header Missing	CWE-918: Server-Side Request Forgery (SSRF)

S.No	Vulnerability Name	CWE Number
5	Drupal Ajax API shows JSONP not disabled by default	CWE-937,1035: Using Components with Known Vulnerabilities
6	Drupal API Vulnerability	CWE-20: Improper Input Validation
7	Improper API Filtration	CWE-79: Cross-site Scripting (XSS)
8	Denial of Service	CWE-400: Uncontrolled Resource Consumption
9	TLS Version 1.0 Protocol Detection	CWE-327: Use of a Broken or Risky Cryptographic Algorithm
10	web.config File Information Disclosure	CWE-497: Exposure of Sensitive System Information

Main Website

S.No	Vulnerability Name	CWE Number
1	SL Medium Strength Cipher Suites Supported (SWEET32)	CWE-200: Exposure of Sensitive Information
2	Cookie SERVERID created without secure flag	CWE-79: Cross-site Scripting (XSS)
3	X-Content-Type Options Header Missing	CWE-918: Server-Side Request Forgery (SSRF)
4	Denial of Service	CWE-400: Uncontrolled Resource Consumption
5	OS Identification	CWE-200: Exposure of Sensitive Information

Detailed Vulnerability Reports: Practice Website

1) Vulnerability Name: Improper Sanitization of File Paths

- **CWE-284:** Improper Access Control
- **OWASP Category:** A01:2021 – Broken Access Control
- **Description:** The application fails to properly sanitize file paths provided by users. This can allow an attacker to traverse the directory structure and access files or directories that should be restricted, such as configuration files or sensitive data.
- **Business Impact:** Unauthorized access to sensitive files can lead to information disclosure, exposure of application logic, and potential system compromise. This can result in data breaches, regulatory penalties, and a loss of customer trust.
- **Steps to Reproduce:**
    1. Identify a file download or inclusion functionality in the web application.
    2. Manipulate the file path parameter by inserting path traversal sequences like ../.
    3. Attempt to access known system files (e.g., ../../../../etc/passwd on Linux or ..\..\..\windows\win.ini on Windows).
    4. If the file content is returned, the vulnerability is confirmed.
- **Recommendation:**
    1. Apply security updates from Drupal as soon as they become available.
    2. Apply the principle of least privilege to web server file permissions.
    3. Conduct regular security audits and vulnerability assessments.
    4. Implement strong access controls to restrict file access to authorized users only.
    5. Minimize the amount of sensitive data stored on the web server.

2) Vulnerability Name: Invalid Certificate

- **CWE-395:** Improper Certificate Validation
- **OWASP Category:** A07:2021 - Identification and Authentication Failures
- **Description:** The service uses an invalid, expired, or self-signed digital certificate, or its certificate chain cannot be properly validated against a trusted Certificate Authority (CA). This undermines the trust model of TLS/SSL.
- **Business Impact:** Improper certificate validation enables man-in-the-middle attacks, data breaches, and malware distribution. Attackers can intercept communication, gain unauthorized access to sensitive information, and compromise system integrity, leading to financial losses and reputational damage.

- **Steps to Reproduce:** Use online tools like SSL Labs (https://www.ssllabs.com/ssltest/analyze?d=www.itsecgames.com) or OpenSSL to inspect the certificate chain, validity dates, and issuer.
- **Recommendation:**
    1. Use digital certificates issued by well-known and trusted Certificate Authorities.
    2. Verify the entire certificate chain, including intermediate certificates.
    3. Check certificate revocation status using CRL or OCSP.
    4. Implement certificate pinning where appropriate.
    5. Keep all software and security protocols updated with the latest patches.

3) Vulnerability Name: Missing X-Frame-Options Header

- **CWE-1021:** Improper Restriction of Rendered UI Layers or Frames
- **OWASP Category:** A04:2021 - Insecure Design
- **Description:** The web application does not include the X-Frame-Options header in its HTTP responses. This header controls whether a page can be displayed in a frame (<frame>, <iframe>, or <object>).
- **Business Impact:** The absence of this header leaves the application vulnerable to clickjacking attacks. Attackers can trick users into interacting with the application unintentionally, leading to data leakage, fraud, user distrust, and reputational damage.
- **Vulnerability Path:** http://www.itsecgames.com/
- **Steps to Reproduce:**
    1. Open the practice website in a browser with developer tools.
    2. Navigate to the Network tab and refresh the page (F5 or Ctrl+R).
    3. Examine the HTML document response headers.
    4. If x-frame-options is not present in the response headers, the vulnerability exists.
- **Recommendation:**
    1. Implement proper UI layer access controls based on user privileges.
    2. Ensure robust session management and user authentication mechanisms.
    3. Use Content Security Policy (CSP) headers to control resource loading and mitigate frame-based attacks.
    4. Keep all software and libraries updated with the latest security patches.

4) Vulnerability Name: X-Content-Type Options Header Missing

- **CWE-918:** Server-Side Request Forgery (SSRF)
- **OWASP Category:** A10:2021 - Server-Side Request Forgery (SSRF)
- **Description:** The X-Content-Type-Options: nosniff header is missing. This header prevents browsers from MIME-sniffing a response away from the declared Content-Type.
- **Business Impact:** Missing this header can lead to data exposure as attackers might force the server to render sensitive data as HTML. It increases security risks by allowing content type manipulation, potentially leading to breaches, data leaks, and unauthorized access, which can harm the organization's reputation.
- **Vulnerability Path:** https://securityheaders.com/?q=http%3A%2F%2Fwww.itecgames.com%2F&followRedirects=on
- **Steps to Reproduce:** Use an online header checker or intercept a response with a proxy like Burp Suite. If the X-Content-Type-Options header is not present in the response, the vulnerability exists.
- **Recommendation:**
    - Configure the X-Content-Type-Options: nosniff header in your web server or application.
    - Implement proper input validation, access controls, and allow-list-based URL validation to mitigate SSRF vulnerabilities.
    - Implement other security headers like X-Frame-Options and Content-Security-Policy for comprehensive protection.

5) Vulnerability Name: Drupal Ajax API shows JSONP not disabled by default

- **CWE-937,1035:** Using Components with Known Vulnerabilities
- **OWASP Category:** A06:2021 - Vulnerable and Outdated Components
- **Description:** This specific vulnerability (CVE-2020-13666) arises because the Drupal AJAX API does not disable JSONP by default. JSONP can be exploited to bypass Same-Origin Policy and perform Cross-site scripting (XSS) attacks.
- **Business Impact:** Using vulnerable components allows attackers to gain unauthorized access, inject malicious code, steal sensitive data, or disrupt system operations. This can result in compromised user data, damaged reputation, legal consequences, and financial losses.
- **Vulnerability Path:** https://www.shodan.io/host/31.3.96.40

- **Steps to Reproduce:** CVE-2020-13666 affects Drupal Core versions prior to 7.73, 8.8.10, 8.9.6, and 9.0.6. Version detection can confirm vulnerability.
- **Recommendation:**
    1. Keep all components, libraries, and dependencies updated with the latest security patches.
    2. Conduct periodic vulnerability assessments to identify potential issues.
    3. Implement a strategy for managing third-party dependencies with continuous security monitoring.
    4. Perform regular security testing, including penetration testing and code reviews.
    5. Maintain a change management process to assess the impact of new components.

6) Vulnerability Name: Drupal API Vulnerability

- **CWE-20:** Improper Input Validation
- **OWASP Category:** A03:2021 - Injection
- **Description:** The software does not validate or incorrectly validates input, which could allow an attacker to inject malicious data or commands. In the context of CVE-2022-25271, certain contributed or custom modules' forms may be vulnerable, allowing attackers to inject disallowed values or overwrite data.
- **Business Impact:** Successful exploitation can lead to unauthorized access to sensitive information, data theft, financial losses, damage to reputation, and legal consequences. It can result in compliance violations, customer trust erosion, and operational disruption.
- **Vulnerability Path:** https://www.shodan.io/host/31.3.96.40
- **Steps to Reproduce:** CVE-2022-25271 affects Drupal core's form API, where certain contributed or custom modules' forms may be vulnerable to improper input validation.
- **Recommendation:**
    1. Implement rigorous input validation on all user inputs.
    2. Use parameterized queries or prepared statements to prevent SQL injection.
    3. Escape user input before incorporating it into SQL queries if parameterized queries are not feasible.
    4. Apply the principle of least privilege to database accounts.
    5. Employ Web Application Firewalls (WAF) to filter malicious requests.
    6. Conduct regular security assessments, including penetration testing.

7) Vulnerability Name: Improper API Filtration

- **CWE-79:** Improper Neutralization of Input During Web Page Generation (Cross-site Scripting)
- **OWASP Category:** A03:2021 – Injection
- **Description:** The product does not neutralize or incorrectly neutralizes user-controllable input before it is placed in output that is used as a web page that is served to other users.
- **Business Impact:** XSS allows attackers to inject malicious scripts into web pages viewed by other users. This can lead to data theft, unauthorized access, compromised user accounts, damaged reputation, and financial losses. Attackers can steal credentials, personal data, or financial details, leading to regulatory violations and legal liabilities.
- **Vulnerability Path:** https://www.shodan.io/host/31.3.96.40
- **Steps to Reproduce:** CVE-2021-326 affects Drupal Core versions prior to 9.1.7, 9.0.12, 8.9.14, and 7.80, where the sanitization API fails to properly filter XSS under certain circumstances.
- **Recommendation:**
    1. Update Drupal and all related modules with the latest security patches.
    2. Implement strong input validation and output encoding to sanitize user inputs.
    3. Implement Content Security Policy (CSP) headers to restrict content sources.
    4. Train developers in secure coding practices for input validation and output escaping.
    5. Conduct regular security scans and penetration tests to identify XSS vulnerabilities.

8) Vulnerability Name: Denial of Service

- **CWE-400:** Uncontrolled Resource Consumption
- **Description:** A Denial of Service (DoS) attack focuses on making a resource unavailable for its intended purpose by overwhelming it with excessive requests, exploiting programming vulnerabilities, or mishandling resources.
- **Business Impact:** Successful DoS attacks cause system slowdowns or complete unresponsiveness, depleting critical resources like CPU, memory, and network bandwidth. This disrupts business productivity, causes financial instability, damages reputation, erodes customer trust, and may lead to regulatory compliance issues and data loss in severe cases.
- **Steps to Reproduce:** Exploit open ports (e.g., 80, 443) to perform SQL injections, cross-site request forgeries, or coordinate DDoS attacks.

- **Recommendation:**
    - Employ a robust Web Application Firewall (WAF) to filter and mitigate suspicious traffic targeting open ports.
    - Configure the WAF to block excessive requests or patterns indicative of an attack.
    - Utilize load balancing techniques to distribute traffic across multiple servers.
    - Implement rate limiting and throttling mechanisms to restrict requests from a single IP address.
    - Monitor traffic patterns and set up alerts for unusual spikes in activity.
    - Keep software and systems updated with the latest security patches.

9) Vulnerability Name: TLS Version 1.0 Protocol Detection

- **CWE-327:** Use of a Broken or Risky Cryptographic Algorithm
- **OWASP Category:** A02:2021 - Cryptographic Failures
- **Description:** The remote service accepts connections encrypted using TLS 1.0. TLS 1.0 has numerous cryptographic design flaws, and while modern implementations mitigate some problems, newer versions like TLS 1.2 and 1.3 are designed to be more secure.
- **Business Impact:** Using TLS 1.0 exposes sensitive data to potential security threats, jeopardizing the integrity of customer information and payment details. This can lead to legal consequences, loss of customer trust, and decreased revenue as security-conscious users avoid the site.
- **Steps to Reproduce:**
    1. Open a terminal or command prompt.
    2. Execute the command: openssl s_client -connect 31.3.96.40:443 -tls1
    3. If the connection is successful, the server supports TLS 1.0.
- **Recommendation:**
    1. Upgrade to secure TLS versions such as TLS 1.2 or TLS 1.3.
    2. Regularly update SSL/TLS configuration to align with security best practices.
    3. Use strong cipher suites and secure configurations while disabling weak ciphers.
    4. Ensure compliance with data protection regulations and industry standards (GDPR, PCI DSS).

10) Vulnerability Name: web.config File Information Disclosure

- **CWE-497:** Exposure of Sensitive System Information to an Unauthorized Control Sphere
- **OWASP Category:** A01:2021 – Broken Access Control
- **Description:** The web.config file, which contains sensitive configuration settings for an ASP.NET application, is accessible via a direct HTTP request.
- **Business Impact:** Exposure of configuration details, including database connection strings and encryption keys, can lead to unauthorized access, data breaches, and system compromise. This results in financial losses, reputational damage, and legal liabilities.
- **Steps to Reproduce:**

```text
GET /web.config HTTP/1.1
Host: web.mmebvba.com
Accept-Charset: iso-8859-1,utf-8;q=0.9,*;q=0.1
Accept-Language: en
Connection: Keep-Alive
User-Agent: Mozilla/4.0 (compatible; MSIE 8.0; Windows NT 5.1; Trident/4.0)
Pragma: no-cache
Accept: image/gif, image/x-xbitmap, image/jpeg, image/pjpeg, image/png, */*
```

- **Recommendation:**
  - Implement proper access controls to restrict access to the web.config file.
  - Prevent direct access to the web.config file through the web server.
  - Encrypt sensitive data within configuration files.
  - Regularly review and update the web.config file, removing unnecessary sensitive information.
  - Ensure the web.config file is properly secured and not accessible to unauthorized users.

Detailed Vulnerability Reports: Main Website

1) Vulnerability Name: SL Medium Strength Cipher Suites Supported (SWEET32)

- **CWE-200:** Exposure of Sensitive Information to an Unauthorized Actor
- **OWASP Category:** A02:2021 – Cryptographic Failures
- **Description:** The remote host supports SSL ciphers offering medium strength encryption (key lengths between 64 and 112 bits, or using the 3DES encryption suite). This is considerably easier to circumvent if the attacker is on the same physical network.
- **Business Impact:** The SWEET32 (Birthday Attack) vulnerability allows attackers to potentially intercept or manipulate encrypted communications. This could lead to unauthorized access, data theft, or injection of malicious code, risking the confidentiality and integrity of sensitive information.
- **Vulnerability Path:** The Sweet32 vulnerability is typically reported with a CVSS score of 7.5.
- **Steps to Reproduce:** Use vulnerability scanners like Nessus or Qualys to detect supported cipher suites.
- **Recommendation:**
    - Update and configure cryptographic libraries to disable support for cipher suites with 64-bit block sizes.
    - Ensure only modern and secure cipher suites are enabled in TLS/SSL configurations.
    - Regularly monitor systems for outdated protocols and promptly apply security patches.
    - Implement strong, unique encryption keys and employ forward secrecy protocols.
    - Use intrusion detection systems and perform regular security audits.

2) Vulnerability Name: Cookie SERVERID created without secure flag

- **CWE-79:** Improper Neutralization of Input During Web Page Generation
- **OWASP Category:** A05:2021 – Security Misconfiguration
- **Description:** A cookie named "SERVERID" is being set without the "Secure" flag. This means the cookie will be transmitted over unencrypted HTTP connections, making it susceptible to network sniffing and session hijacking.
- **Business Impact:** This vulnerability can lead to reputational damage, compromised customer trust, and legal repercussions. Attackers can exploit this to hijack user sessions, leading to data breaches, fraud, and financial losses.

- **Vulnerability Path:** https://securityheaders.com/?q=http%3A%2F%2Fvtop.vit.ac.in%2F&followRedirects=on
- **Steps to Reproduce:** Use a browser's developer tools or an intercepting proxy. Observe the Set-Cookie header in the HTTP response. If the cookie does not have the ; Secure flag, this vulnerability is present.
- **Recommendation:**
    1. Update and patch all systems with the latest security updates.
    2. Implement strong input validation and output encoding.
    3. Implement Content Security Policy (CSP) headers.
    4. Train developers in secure coding practices.
    5. Conduct regular security scans and penetration tests.

3) Vulnerability Name: X-Content-Type Options Header Missing

- **CWE-918:** Server-Side Request Forgery (SSRF)
- **OWASP Category:** A10:2021 - Server-Side Request Forgery (SSRF)
- **Description:** The X-Content-Type-Options: nosniff header is missing.
- **Business Impact:** The header missing impacts data exposure (attackers might force the server to render sensitive data as HTML), increases security risks, and can harm the organization's reputation and trustworthiness.
- **Vulnerability Path:** https://securityheaders.com/?q=http%3A%2F%2Fvtop.vit.ac.in%2F&followRedirects=on
- **Steps to Reproduce:** Use an online header checker or intercept a response with a proxy. If the X-Content-Type-Options header is not present, the vulnerability exists.
- **Recommendation:**
    - Implement X-Content-Type-Options: nosniff header.
    - Ensure proper input validation, access controls, and allow-list-based URL validation.
    - Implement other security headers like X-Frame-Options and Content-Security-Policy.

4) Vulnerability Name: Denial of Service

- **CWE-400:** Uncontrolled Resource Consumption
- **Description:** A Denial of Service (DoS) attack focuses on making a resource unavailable for its intended purpose by overwhelming it with excessive requests.
- **Business Impact:** DoS attacks cause system slowdowns or unresponsiveness, disrupting business productivity, causing financial instability, damaging reputation, eroding customer trust, and potentially leading to data loss and regulatory compliance issues.
- **Steps to Reproduce:** Exploit open ports 443 to perform SQL injections, cross-site request forgeries, and DDoS attacks.
- **Recommendation:**
    - Employ a robust Web Application Firewall (WAF).
    - Configure WAF to block excessive requests or attack patterns.
    - Utilize load balancing techniques.
    - Implement rate limiting and throttling mechanisms.
    - Monitor traffic patterns and set up alerts for unusual spikes.
    - Keep software and systems updated with security patches.

5) Vulnerability Name: OS Identification

- **CWE-200:** Exposure of Sensitive Information to an Unauthorized Actor
- **OWASP Category:** A02:2021 - Cryptographic Failures
- **Description:** Using a combination of remote probes (TCP/IP, SMB, HTTP, etc.), it is possible to guess the name and version of the remote operating system in use.
- **Business Impact:** Exposure of sensitive information can lead to financial losses, regulatory penalties, and reputational damage. Customers may lose trust, legal consequences may arise, and intellectual property theft can compromise competitive advantage.
- **Steps to Reproduce:** Use nmap -O <TARGET> to enable OS detection.
- **Recommendation:**
    - Implement robust encryption protocols for data in transit and at rest.
    - Enforce stringent access controls and strong authentication measures.
    - Conduct regular vulnerability assessments and penetration testing.
    - Provide comprehensive security awareness training to employees.

- Employ data masking techniques for non-production environments.
- Utilize Data Loss Prevention (DLP) solutions.
- Maintain compliance with data protection laws and regulations.
- Establish a well-defined incident response plan.

Stage 2: Vulnerability Assessment with Nessus

Overview

Nessus is a vulnerability scanner developed by Tenable, Inc. It is a remote scanner, meaning it does not need to be installed on the target host to scan it. Nessus works by sending a variety of probes to the target host in order to identify vulnerabilities. These probes can include network traffic, operating system commands, and application requests.

Nessus uses a plugin-based architecture. Plugins are small pieces of code designed to identify specific vulnerabilities. There are over 1200 plugins available for Nessus, covering a wide range of vulnerabilities in operating systems, applications, and network devices.

When Nessus scans a host, it first uses a discovery plugin to identify the operating system and services running on the host. Once the host has been discovered, Nessus runs a series of plugins to test for vulnerabilities. For each vulnerability found, it provides a detailed report that includes the severity, steps to reproduce, and steps to remediate.

Scanning Process:

1. Nessus connects to the target host using various protocols (TCP, UDP, ICMP).
2. Nessus sends probes to identify the operating system and services.
3. Nessus runs plugins to test for vulnerabilities.
4. Nessus generates a comprehensive report.

Benefits of Nessus:

- Comprehensive vulnerability coverage with over 1200 plugins
- User-friendly interface for easy configuration and scanning
- Detailed reporting with severity levels and remediation steps
- Scalability for scanning single hosts, networks, or web applications
- Flexibility for scheduled or manual scans

Target Website: http://testfire.net/
Target IP Address: 65.61.137.117

Nessus Scan Results

S.No	Vulnerability Name	Severity	Plugin ID
1	TLS Version 1.0 Protocol Detection	MEDIUM	104743
2	TLS Version 1.1 Protocol Deprecated	MEDIUM	157288
3	SSL/TLS Diffie-Hellman Modulus ≤ 1024 Bits (Logjam)	LOW	83875
4	ICMP Timestamp Request Remote Date Disclosure	INFO	101145
5	Additional DNS Hostnames	INFO	46180
6	Apache Tomcat Detection	INFO	39446
7	Common Platform Enumeration (CPE)	INFO	45590
8	Device Type	INFO	54615
9	HSTS Missing From HTTPS Server	INFO	84502
10	HTTP Server Type and Version	INFO	10107

Detailed Nessus Vulnerability Reports

1) Vulnerability Name: TLS Version 1.0 Protocol Detection

- **Plugin ID:** 104743
- **Severity:** MEDIUM
- **Port:** 443
- **Description:** The remote service accepts connections encrypted using TLS 1.0. TLS 1.0 has numerous cryptographic design flaws. Modern implementations mitigate some problems, but newer versions like TLS 1.2 and 1.3 are designed against these flaws and should be used whenever possible. PCI DSS v3.2 requires that TLS 1.0 be disabled

entirely by June 30, 2018, except for POS POI terminals verified as not susceptible to known exploits.

- **Solution:** Enable support for TLS 1.2 and 1.3, and disable support for TLS 1.0.
- **Business Impact:** Continued use of TLS 1.0 exposes the organization to heightened security risks, increasing the likelihood of data breaches and cyberattacks. Non-compliance with PCI DSS v3.2 risks regulatory penalties and erodes customer trust.

2) Vulnerability Name: TLS Version 1.1 Protocol Deprecated

- **Plugin ID:** 157288
- **Severity:** MEDIUM
- **Port:** 443
- **Description:** The remote service accepts connections encrypted using TLS 1.1. TLS 1.1 lacks support for current and recommended cipher suites. Ciphers that support encryption before MAC computation and authenticated encryption modes such as GCM cannot be used with TLS 1.1. As of March 31, 2020, endpoints not enabled for TLS 1.2 and higher will no longer function properly with major web browsers and vendors.
- **Solution:** Enable support for TLS 1.2 and/or 1.3, and disable support for TLS 1.1.

3) Vulnerability Name: SSL/TLS Diffie-Hellman Modulus ≤ 1024 Bits (Logjam)

- **Plugin ID:** 83875
- **Severity:** LOW
- **Port:** 443
- **Description:** The remote host allows SSL/TLS connections with one or more Diffie-Hellman moduli less than or equal to 1024 bits. Through cryptanalysis, a third party may be able to find the shared secret in a short amount of time (depending on modulus size and attacker resources). This may allow an attacker to recover the plaintext or potentially violate the integrity of connections.
- **Solution:** Reconfigure the service to use a unique Diffie-Hellman moduli of 2048 bits or greater.
- **Business Impact:** While severity is low, the risk of data breaches and integrity compromise should not be underestimated. Addressing this vulnerability is essential to maintain data security and user trust.

4) Vulnerability Name: ICMP Timestamp Request Remote Date Disclosure

- **Plugin ID:** 101145
- **Severity:** INFO
- **Port:** 0

- **Description:** The remote host answers to an ICMP timestamp request. This allows an attacker to know the date set on the targeted machine, which may assist an unauthenticated, remote attacker in defeating time-based authentication protocols.
- **Solution:** Filter out ICMP timestamp requests (13) and outgoing ICMP timestamp replies (14).
- **Business Impact:** This does not pose an immediate threat but could be used by attackers for reconnaissance purposes. It should be addressed as part of a broader security posture.

5) Vulnerability Name: Additional DNS Hostnames

- **Plugin ID:** 46180
- **Severity:** INFO
- **Port:** 0
- **Description:** Hostnames different from the current hostname have been collected by miscellaneous plugins. Nessus has generated a list of hostnames that point to the remote host. Note that these are only the alternate hostnames for hosts discovered on a web server. Different web servers may be hosted on name-based virtual hosts.
- **Solution:** If you want to test them, re-scan using the special vhost syntax, such as: www.example.com [192.0.32.10]
- **Business Impact:** This does not pose an immediate threat but can be useful for understanding server configuration and infrastructure.

6) Vulnerability Name: Apache Tomcat Detection

- **Plugin ID:** 39446
- **Severity:** INFO
- **Port:** 80
- **Description:** Nessus was able to detect a remote Apache Tomcat web server.
- **Solution:** N/A
- **Business Impact:** This informational finding identifies the presence of an Apache Tomcat server. While not a security threat, it helps network administrators understand technologies in use on their network.

7) Vulnerability Name: Common Platform Enumeration (CPE)

- **Plugin ID:** 45590
- **Severity:** INFO
- **Port:** 0

- **Description:** By using information obtained from a Nessus scan, this plugin reports CPE matches for various hardware and software products found on a host. If an official CPE is not available, this plugin computes the best possible CPE based on available information.
- **Solution:** N/A
- **Business Impact:** This provides valuable insights for asset management and inventory purposes, aiding administrators in understanding components within their network.

8) Vulnerability Name: Device Type

- **Plugin ID:** 54615
- **Severity:** INFO
- **Port:** 0
- **Description:** Based on the remote operating system, it is possible to determine the remote system type (e.g., printer, router, general-purpose computer).
- **Solution:** N/A
- **Business Impact:** This offers valuable insights for network administrators to better understand and classify devices in their network, aiding in network management and optimization.

9) Vulnerability Name: HSTS Missing From HTTPS Server

- **Plugin ID:** 84502
- **Severity:** INFO
- **Port:** 443
- **Description:** The remote HTTPS server is not enforcing HTTP Strict Transport Security (HSTS). HSTS is an optional response header that instructs browsers to communicate only via HTTPS. The lack of HSTS allows downgrade attacks, SSL-stripping man-in-the-middle attacks, and weakens cookie-hijacking protections.
- **Solution:** Configure the remote web server to use HSTS.
- **Business Impact:** The absence of HSTS potentially exposes the server to downgrade attacks and weakens protections against cookie hijacking, posing risks to data confidentiality and user security.

10) Vulnerability Name: HTTP Server Type and Version

- **Plugin ID:** 10107
- **Severity:** INFO
- **Port:** 443

- **Description:** This plugin attempts to determine the type and version of the remote web server.
- **Solution:** N/A
- **Business Impact:** While not an immediate security threat, this information can aid attackers in exploiting known vulnerabilities specific to that software. Restricting such information enhances the server's overall security posture.

Stage 3: Zero Trust Network Access (ZTNA) Implementation with Zscaler

Title: Implementing Zero Trust Network Access (ZTNA) with Zscaler: Securing User-to-Application Connectivity

Understanding Zero Trust Network Access (ZTNA)

Zero Trust Network Access (ZTNA) is an IT security framework that provides secure remote access to an organization's internal applications and data based on clearly defined access control policies. Unlike traditional Virtual Private Networks (VPNs), which grant users broad access to the entire network, ZTNA operates on a **"never trust, always verify"** principle. It grants users access only to the specific applications they need, and only after their identity and device have been verified.

ZTNA is a critical component of modern security architecture. It helps organizations move away from the concept of a trusted internal network and an untrusted external network (the "castle-and-moat" model). By hiding the internal network from the internet and establishing one-to-one, encrypted connections between users and applications, ZTNA drastically reduces the attack surface and stops common threats like lateral movement by attackers.

The ZTNA Access Cycle

The ZTNA cycle is a continuous process of verifying identity, establishing secure connections, and enforcing policy for every access request. It typically consists of the following steps:

1. **User Request:** A user, from any location (office, home, coffee shop), requests access to a specific internal application (e.g., an HR portal, a finance server, or an internal code repository).
2. **Identity and Device Verification:** Before any connection is established, the ZTNA solution verifies who the user is (via multi-factor authentication or MFA) and checks the security posture of their device. This includes ensuring the device has up-to-date antivirus, a healthy firewall, and is not compromised.

3. **Policy Enforcement:** The ZTNA solution checks the access policy. Does this specific user, with this specific device, have permission to access the requested application at this specific time?
4. **Establish Secure Connection:** If the policy allows access, the ZTNA solution creates a highly secure, encrypted micro-tunnel between the user's device and the target application. Crucially, the user never gains access to the underlying network, only to the application itself.
5. **Continuous Monitoring:** The ZTNA solution continues to monitor the connection for anomalies. If the user's behavior becomes suspicious or their device posture changes, the connection can be terminated in real-time.
6. **Logging and Analytics:** All access attempts, successful and failed, are logged and analyzed. This data is used for auditing, threat hunting, and refining access policies.

Zscaler: A Cloud-Native Security Platform

Zscaler is a cloud-native security platform that delivers its services entirely from the cloud, eliminating the need for on-premises hardware like VPN concentrators. For ZTNA, Zscaler offers **Zscaler Private Access (ZPA)** . ZPA is a service that provides seamless, identity-based access to private applications without ever placing users on the network.

Zscaler ZPA operates on a simple principle: the application is "inside" the Zscaler cloud, and the user connects to it there. It collects data from a variety of sources, including:

1. **Identity Providers (IdP):** Such as Azure AD, Okta, or Ping Identity, to verify user identities.
2. **Device Posture Partners:** Such as CrowdStrike, Tanium, or Microsoft Defender for Endpoint, to check the health of the endpoint.
3. **Business Applications:** The internal web apps, file servers, and legacy applications hosted in data centers or public clouds (e.g., AWS, Azure).
4. **Zscaler App Connectors:** Lightweight software components deployed in the data center or cloud that initiate outbound connections to the Zscaler cloud, making internal servers invisible to the internet.

Key Features of Zscaler ZPA:

- **Identity-Based Access:** Policies are built around user and group identities, not IP addresses.
- **Application Segmentation:** Users are granted access to specific applications, not entire subnets.
- **Least Privilege Access:** Ensures users have the minimum level of access required to perform their jobs.

The Zscaler ZTNA Cycle

The Zscaler ZTNA cycle is the practical application of the ZTNA principles using the Zscaler cloud. The cycle typically consists of the following steps:

1. **User Login and Authentication:** A user attempts to access an internal application. They are redirected to their organization's Identity Provider (IdP) for authentication and MFA.
2. **Device Posture Check (Integration):** Zscaler checks with integrated endpoint protection tools to ensure the device is compliant with security policies.
3. **Policy Lookup and Decision:** The Zscaler cloud evaluates the request against the defined access policies. It verifies that the user, device, and context (time, location) are authorized for that specific application.
4. **Establishing the Micro-tunnel:** Once authorized, the Zscaler cloud instructs the user's Zscaler Client Connector and the appropriate Zscaler App Connector in the data center to establish an encrypted connection. The user is now connected to the application.
5. **Traffic Inspection (Optional):** For sanctioned applications, Zscaler can optionally inspect the traffic for data loss prevention (DLP) or to detect malware.
6. **Logging and Analytics:** All activity is logged in the Zscaler cloud portal for reporting, threat hunting, and compliance.

MISP (Malware Information Sharing Platform) Integration

While Zscaler is a primary access control tool, it can be enhanced by integrating with threat intelligence platforms like MISP.

1. **Threat Intelligence Feed:** MISP can aggregate and share threat intelligence about malicious domains, IPs, and URLs. This list of indicators of compromise (IOCs) can be fed into Zscaler's cloud.
2. **Proactive Blocking:** If a user, even one authorized by ZPA, attempts to access a known malicious site that has been flagged by the MISP feed, Zscaler Internet Access (ZIA) can block the request, preventing a potential infection.
3. **Enhanced Visibility:** Security teams can use MISP to cross-reference Zscaler logs. If a user's behavior is anomalous, the team can check MISP to see if the destination IP or domain is associated with a known threat actor or campaign.

College Network Information (Example: VIT)

The college network is a complex environment serving thousands of students, faculty, and staff, with both on-premises and remote access requirements. It consists of various network

segments, including academic, research, administrative, and guest networks, each with different security requirements. Critical assets include student records, research data, library systems, and administrative portals.

The need for ZTNA arises from the limitations of the current VPN model:

- **Over-permissioned Access:** Once on the VPN, a faculty member's compromised laptop could potentially scan and attack the entire internal network (lateral movement).
- **Hidden Applications:** Some research applications need to be accessed by external collaborators without granting them full network access.
- **User Experience:** The legacy VPN can be slow and cumbersome, often requiring users to route all their internet traffic through the university.

Deployment Plan for Zscaler ZTNA in a College Environment

To deploy Zscaler ZTNA (ZPA) in a college, the approach would focus on identifying specific applications that need to be secured and the user groups that need access to them. This is a shift from securing the network to securing access to applications.

The deployment plan would include the following steps:

1. **Application Discovery:** Catalog all internal applications (e.g., ERP, library system, research servers, file shares) and identify which ones are currently exposed via the VPN.
2. **User Segmentation:** Define user groups (e.g., Students, Faculty - Research, Faculty - Admin, IT Staff, External Collaborators) and the specific applications each group needs.
3. **Deploying Zscaler App Connectors:** Install lightweight App Connector VMs in the university's data center. These connectors will establish outbound connections to the Zscaler cloud, making the application servers invisible to the public internet.
4. **Integrating Identity Provider:** Connect Zscaler to the university's existing Identity Provider (e.g., Microsoft Azure AD) to leverage current usernames, passwords, and MFA policies.
5. **Configuring Access Policies:** Create granular policies in the Zscaler admin portal. For example: "The 'Faculty' group can access the 'ERP Application' from any device that has antivirus enabled, after MFA."
6. **Deploying Zscaler Client Connector:** Students, faculty, and staff will install the Zscaler Client Connector on their laptops and mobile devices. This small client intelligently routes traffic for private applications through ZPA and public traffic through ZIA.

Threat Intelligence Integration

Effective security relies on consuming and acting upon threat intelligence from multiple sources:

1. **Zscaler Cloud Logs:** Internal logs provide real-time data on who is accessing which applications, from which locations, and on what devices.
2. **External Threat Feeds (MISP):** Information about new malware campaigns, phishing kits, or command-and-control servers being used by attackers.
3. **Identity Provider Logs:** Anomalies in login attempts, such as a user logging in from two geographically impossible locations in a short time, can signal a compromised account.

This intelligence is used for:

1. **Adaptive Access Control:** If threat intelligence suggests a user's location is high-risk, Zscaler can step-up authentication requirements or block access entirely.
2. **Hunting for Compromised Accounts:** Analysts can use Zscaler logs to see if a user is suddenly accessing applications they never have before, which could be a sign of credential theft.
3. **Blocking Malicious Communication:** Threat intelligence feeds can be used by Zscaler Internet Access (ZIA) to block outbound connections to known malicious sites, even if they originate from a user inside the university.

Incident Response in a ZTNA Model

Incident response in a ZTNA model is faster and more targeted because the environment is inherently more controlled.

1. **Detection:** The SOC team detects an anomaly. For example, a faculty account is downloading an unusually high volume of data from a research server at 3 AM. Zscaler logs provide the exact details: the user, the device, the application, and the amount of data transferred.
2. **Containment:** The SOC team can instantly isolate the user. In the Zscaler portal, they can simply deactivate that user's access or block their specific device. Because there is no network-level access, this action immediately severs all connections to every application, preventing lateral movement.
3. **Eradication:** The compromised device is taken offline for malware scanning and cleaning by the IT helpdesk.
4. **Recovery:** Once the device is clean and the user's password is reset, the Zscaler policy is re-enabled, granting the user access again.

5. **Post-incident review:** The security team analyzes the logs to understand how the account was compromised (e.g., a phishing email) and updates Zscaler policies, perhaps by adding a block for similar malicious domains or tightening access rules for that specific research application.

Benefits of this incident response approach:

1. **Near-Instant Containment:** Actions take seconds in a cloud dashboard, not minutes or hours configuring firewall rules.
2. **No Lateral Movement:** The attacker is contained to the single compromised application, protecting the rest of the university's network.
3. **Granular Forensics:** Zscaler provides a detailed audit trail of every connection, making investigations faster and more accurate.
4. **Compliance:** This model helps demonstrate compliance with data protection regulations by proving that access to sensitive student or research data is tightly controlled and audited.

Understanding Zscaler Private Access (ZPA)

Zscaler Private Access (ZPA) is a cloud service that embodies the ZTNA principle of "connect a user to an app, not the network." Here are the key things to understand:

1. **It is "Application-Centric," Not "Network-Centric":** You define policies based on applications (e.g., "SAP Server 01"), not network constructs like IP subnets. This aligns security with how businesses actually operate.
2. **It Hides the Network:** ZPA uses App Connectors inside your data center that only make outbound connections to the Zscaler cloud. Your internal servers never need to open inbound firewall ports, making them invisible to port scanners and attackers on the internet.
3. **It uses a "Dark Cloud" Approach:** To an external scanner, the organization's applications simply don't exist. They only become visible to authorized users after a multi-step verification process.
4. **It Provides a Superior User Experience:** Users don't have to mess with clunky VPN clients. The Zscaler Client Connector works silently in the background. For web apps, access can even be agentless, requiring only a modern browser.
5. **It is Inherently Scalable:** Because it's a cloud service, ZPA scales instantly to handle thousands of concurrent users, unlike a legacy VPN appliance that needs to be manually upgraded.

Conclusion

In conclusion, exploring the realm of AI for web security and Zero Trust architectures has been an insightful journey through various stages.

In **Stage 1**, understanding web application testing unveiled the critical importance of assessing vulnerabilities and ensuring the security of web-based systems. This foundational knowledge served as a cornerstone for comprehending the subsequent stages.

Stage 2 delved into the intricacies of Nessus reports, emphasizing the significance of detailed vulnerability assessments. Analyzing these reports provided valuable insights into potential threats, enabling informed decision-making for robust security measures.

Stage 3 introduced the pivotal concept of Zero Trust Network Access (ZTNA) and its implementation using Zscaler Private Access (ZPA). It demonstrated a paradigm shift from traditional VPN-based network access to secure, identity-based user-to-application connectivity. The solution showcased the power of hiding internal applications from the internet, enforcing granular access policies, and eliminating lateral movement. This underscores the need for modern, proactive cybersecurity strategies in an era of remote work and cloud adoption.

Considering the future scope, **Stage 1** illuminated the evolving landscape of web application testing. With advancements in AI, there is a promising avenue for more sophisticated testing methodologies, ensuring even higher levels of security for web applications.

Stage 2 highlighted the future trajectory of the testing process, indicating a shift towards more automation and integration of AI-driven tools. This evolution promises quicker, more accurate vulnerability assessments, enhancing overall cybersecurity postures.

Stage 3 illuminated the future potential of Zero Trust Network Access (ZTNA) and Zscaler technologies. With AI integration and machine learning, these systems are poised to become more intelligent and adaptive, enabling real-time risk assessment based on user behavior and device posture. Enhanced automation in policy enforcement, predictive analytics for threat detection, and improved scalability are anticipated, empowering organizations to stay ahead of sophisticated cyber threats in an increasingly distributed world.

In essence, the project not only provided a deep understanding of the current state of Zero Trust using ZSCALER but also offered a glimpse into the exciting future of cybersecurity. Embracing these advancements will be crucial for organizations aiming to fortify their digital landscapes in the face of ever-evolving cyber threats.

Future Scope

Future of Web Application Testing

The future of WAT is bright, as web applications become increasingly complex and sophisticated. Organizations are increasingly reliant on web applications to conduct business, and they need to be confident that their web applications are secure and reliable.

Key trends shaping the future of WAT:

- **Increased focus on security:** Security is a top priority, and WAT is essential for identifying and mitigating security vulnerabilities as web applications become increasingly targeted by cyberattacks.
- **Rise of Artificial Intelligence (AI) and Machine Learning (ML):** AI and ML will automate repetitive tasks like test case generation and execution, and identify complex patterns and anomalies difficult to detect with traditional methods.
- **Shift to cloud-based testing:** Cloud-based WAT platforms offer advantages such as scalability, flexibility, and cost-effectiveness.
- **Increased focus on User Experience (UX) testing:** UX testing ensures web applications are easy to use and meet user needs, becoming increasingly important as organizations recognize the value of positive user experience.

Future Scope of the Testing Process

The future scope of the testing process is bright, as software becomes increasingly complex and sophisticated.

Key trends:

- **Increased focus on continuous testing:** Testing software throughout the SDLC, from development to deployment, helps identify defects early and prevent problems later.
- **Rise of AI and ML:** AI and ML will automate repetitive tasks and identify complex patterns and anomalies.
- **Shift to cloud-based testing:** Cloud-based platforms offer scalability, flexibility, and cost-effectiveness.
- **Increased focus on testing automation:** Automation saves time and resources and improves the quality and efficiency of testing.

Future Scope of SOC / SIEM and ZTNA

The future scope of Zero Trust Network Access (ZTNA) and the broader Zero Trust framework is poised for significant evolution:

1. **AI-Driven Zero Trust (AI-ZTA):** Future implementations will leverage advanced AI and deep learning to move beyond static policies. Systems will continuously learn user behavior, predict anomalous activities before they occur, and dynamically adjust access

privileges in real-time based on contextual risk, creating a truly "adaptive" security posture.

2. **Integration with Edge Computing and IoT:** Zero Trust principles will extend to resource-constrained environments like IoT devices and edge computing nodes. Lightweight Zero Trust agents and protocols will authenticate and secure machine-to-machine (M2M) communication at the edge, preventing large-scale botnet attacks.
3. **Identity-Centric Everything:** Identity will become the ultimate security perimeter. Zero Trust will evolve to incorporate advanced identity verification methods, including continuous behavioral biometrics (keystroke dynamics, gait analysis) and decentralized identity solutions using blockchain, ensuring continuous user verification.
4. **Convergence with SASE Architecture:** Zero Trust will be a core component of the Secure Access Service Edge (SASE) framework. This convergence will provide organizations with a unified, cloud-delivered stack combining ZTNA, Secure Web Gateways (SWG), Cloud Access Security Brokers (CASB), and Firewall-as-a-Service (FWaaS), simplifying management and ensuring consistent security.
5. **Zero Trust for Operational Technology (OT) and Critical Infrastructure:** Sectors like energy, manufacturing, and healthcare will increasingly adopt Zero Trust to protect critical infrastructure. Future scope includes air-gapped Zero Trust solutions and specialized protocols for legacy industrial control systems (ICS) and SCADA environments.
6. **Automated Threat Hunting and Response:** Future Zero Trust platforms will integrate with Security Orchestration, Automation, and Response (SOAR) tools to enable automated threat hunting. Upon detecting a compromised identity or device, the system will not just block access but automatically initiate investigation playbooks, isolate affected systems, and even roll back malicious changes.